Ramp energy lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare gravitational energy change with measured kinetic energy.

SUCCESS CRITERIA - I can...

1. Measure height vertically and speed at a stated location.
2. Calculate an energy balance.
3. Discuss rotation, resistance and uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “rotation” and “thermal energy” to compare your model or result. Sentence starters are optional.

rotation: turning about an axis

thermal energy: internal energy associated with microscopic motion/interactions

energy budget: an account of all energy stored and transferred

01 Warm-up: bring back an earlier idea

A ball is thrown vertically upward at 15 m/s from h=0.
Use g=10 and neglect drag.
Find maximum height by energy.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: mg $\Delta h = \Delta K_{translation} + \Delta K_{rotation} + \Delta E_{thermal}$.
Closed energy accounting near Earth; Δh is the positive drop. Determine whether the object rolls or slides; use the approved ramp setup.

Worked reasoning: A supplied ramp run loses 6 J of gravitational potential energy but gains only 4 J of translational kinetic energy. The 2 J difference needs an account, such as rotation or friction.

Your turn: Draw an energy account with the 2 J remainder explicitly labeled as not yet identified.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

Ramp energy lab

antocciphysics

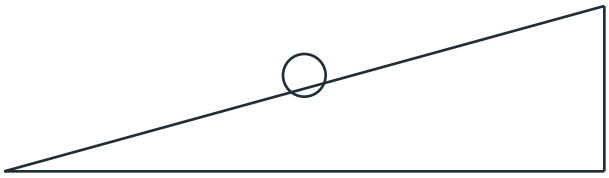
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

On the ramp sketch, mark the vertical height h and the along-ramp path. Which belongs in mgh?

Sentence starter: The ... in my model represents ... . Its label shows ... .



Label h vertically. The marker represents the object on the sloping surface.

Marzano check - circle: ① ② ③

04 Apply the physics

A lab report says a 0.50 kg cart starts from rest, drops 0.20 m vertically and reaches 2.5 m/s.
Use g = 9.8 N/kg.
Compare the gravitational energy available with the reported final kinetic energy. What should the group check before claiming energy was created?

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

05 Explain your reasoning

Create a table for repeated trials. Include actual height, mass, bottom speed,ΔU, K and one observation about losses or rotation.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

At fixed momentum magnitude, mass becomes 4 times as large. Derive the kinetic-energy multiplier using $K=p^2/(2m)$.
Show the energy changes and transfers; label units.

Word help: momentum = mass times velocity, including direction

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

Ramp energy lab

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Supplied data: m=0.50 kg, h=0.40 m, v=2.4 m/s, g=9.8.
Find mgh, K and fraction of initial gravitational energy appearing as translation.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.5 kg cart drops 0.4 m vertically and reaches 2 m/s from rest.
Use g=10.
Find mechanical energy not in translation and average resistive force if all of it is friction over a 1 m path.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3